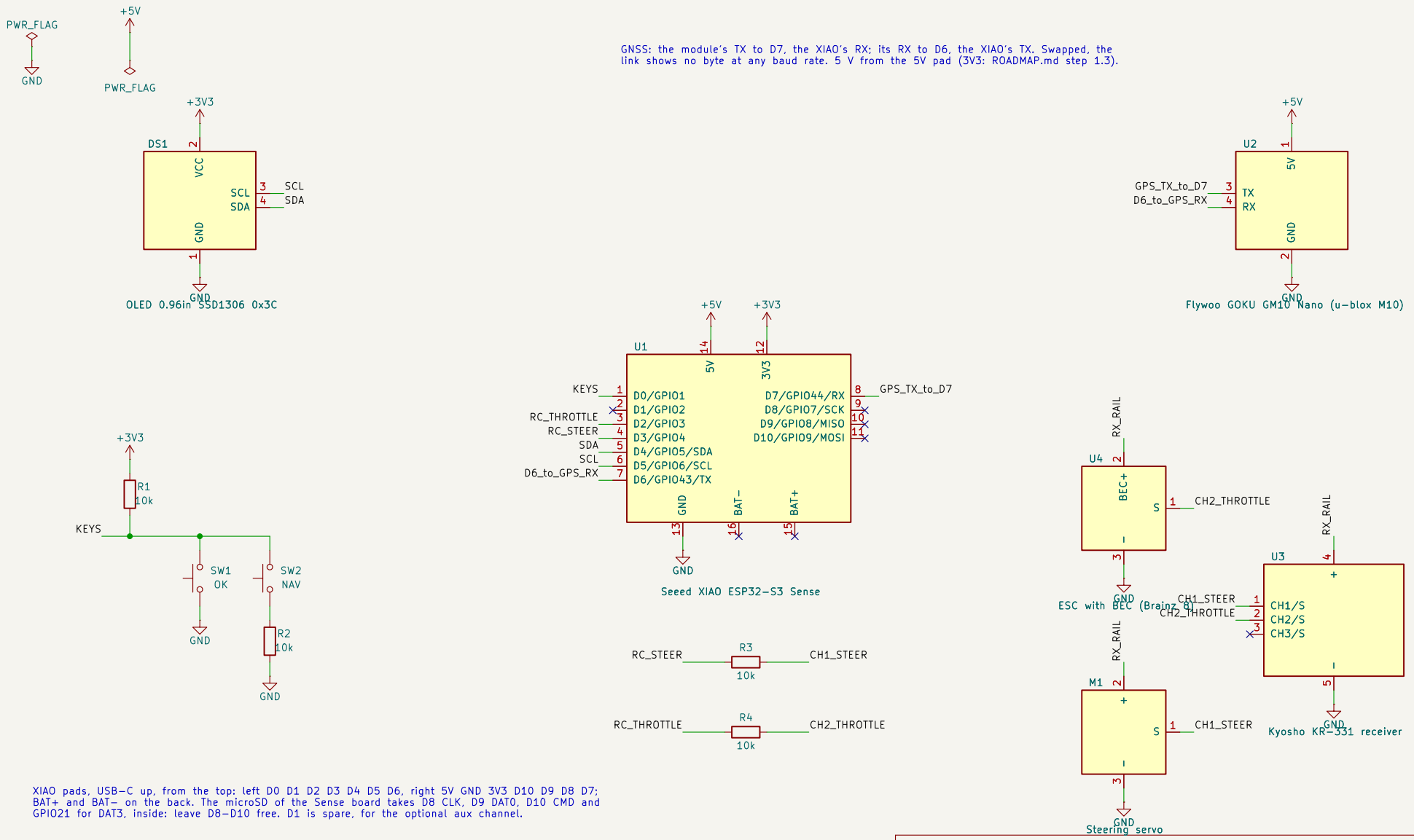

PWR_FLAG
GND
+5V
PWR_FLAG
GNSS: the module's TX to D7, the XIAO's RX; its RX to D6, the XIAO's TX. Swapped, the link shows no byte at any baud rate. 5 V from the 5V pad (3V3: ROADMAP.md step 1.3).
+3V3
DS1
VCC
SCL
SDA
GND
SCL
SDA
OLED 0.96in SSD1306 0x3C
+5V
U2
5V
GPS_TX_to_D7
D6_to_GPS_RX
TX
RX
GND
Flywoo GOKU GM10 Nano (u-blox M10)
+5V
+3V3
U1
5V
3V3
KEYS
RC_THROTTLE
RC_STEER
SDA
SCL
D6_to_GPS_RX
D0/GPIO1
D1/GPIO2
D2/GPIO3
D3/GPIO4
D4/GPIO5/SDA
D5/GPIO6/SCL
D6/GPIO43/TX
D7/GPIO44/RX
D8/GPIO7/SCK
D9/GPIO8/MISO
D10/GPIO9/MOSI
GPS_TX_to_D7
GND
BAT-
BAT+
GND
Seeed XIAO ESP32-S3 Sense
RX_RAIL
U4
BEC+
S
CH2_THROTTLE
GND
ESC with BEC (Brainz G)
RX_RAIL
U3
CH1_STEER
CH2_THROTTLE
CH1/S
CH2/S
CH3/S
GND
Kyosho KR-331 receiver
+3V3
R1
10k
KEYS
SW1
OK
SW2
NAV
GND
R2
10k
GND
RC_STEER
R3
10k
CH1_STEER
RC_THROTTLE
R4
10k
CH2_THROTTLE
RX_RAIL
M1
S
CH1_STEER
GND
Steering servo
XIAO pads, USB-C up, from the top: left D0 D1 D2 D3 D4 D5 D6, right 5V GND 3V3 D10 D9 D8 D7; BAT+ and BAT- on the back. The microSD of the Sense board takes D8 CLK, D9 DAT0, D10 CMD and GPIO21 for DAT3, inside: leave D8-D10 free. D1 is spare, for the optional aux channel.
Keys: the two-button interim ladder (PLAN-ESP32.md 3.1): OK 0 V, NAV 1.62 V, none 3.12 V.
Receiver taps: 10 k in series, enough for the KR-331's 3.3 V logic (tools/rc-scope).
Unknown receiver: tools/rc-scope first, with 10 k from D3 to GND as a divider. CH1 by unplugging the servo or with a Y-cable; CH2 needs a Y-cable, the ESC powering the receiver through it. The receiver's + rail stays off the XIAO, which runs from USB.
As wired on 2026-10-19. Powered from USB-C; the receiver's + rail is not connected.
RC inputs on the bench: 10 k in series, for receivers measured at 3.3 V logic only.
Pin map: PLAN-ESP32.md 2.1. Measurements: docs/hardware.md. The car: car.kicad_sch.
esp32-rc-telemetry
Sheet: /
File: bench.kicad_sch
Title: Bench build: XIAO, display, two keys, GNSS, receiver taps
Size: A4
Date: 2026-10-19
Rev: 1
KiCad E.D.A. 10.0.1
Id: 1/1